

Key words And Terms

special theory of relativity

狭义相对论

transformation

变换

proper time

固有时间

time dilation

时间延迟

proper length

固有长度

length contraction

长度收缩

clock synchronization

时钟同步化

simultaneity

同时性

twin paradox

双生子佯谬

Chapter 21 Special Theory of Relativity

21.1 Galilean Relativity

21.2 Postulates of the special theory of relativity

21.3 Simultaneity; Time Dilation and length Contraction

21.4 Lorentz transformation

21.5 Relativistic mass, momentum and energy

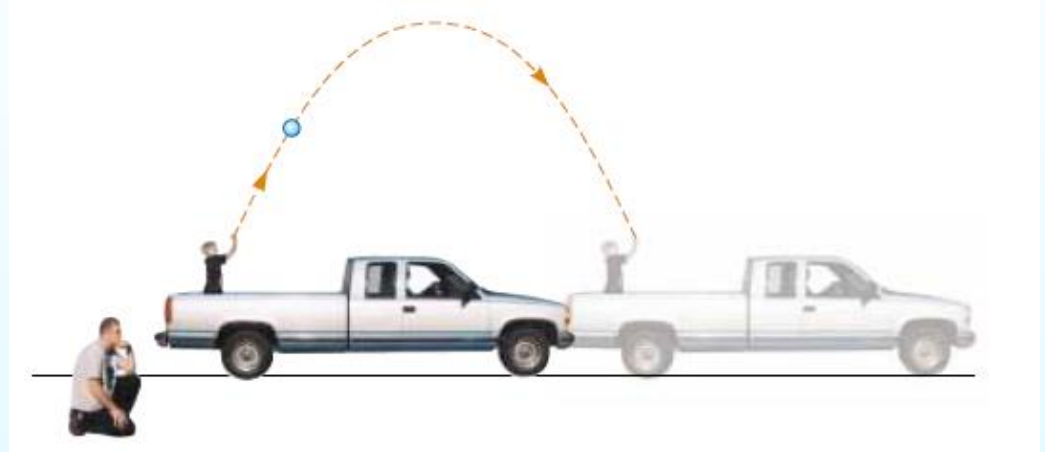
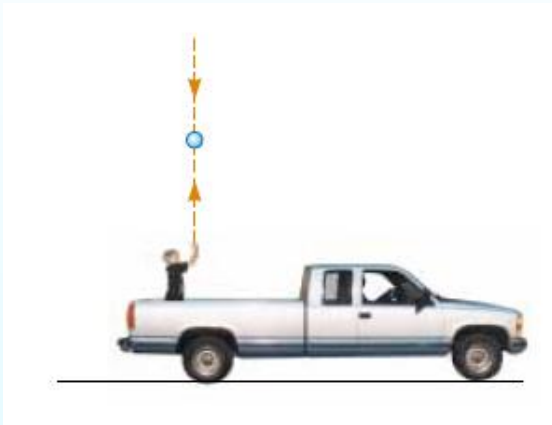
Chapter 21 Special Theory of Relativity

21.1 Galilean Relativity

1. Relativity principle

An **inertial frame** of reference is one in which an object is observed to have **no acceleration** when **no forces** act on it.

The basic laws of physics are the **same** in all inertial reference frames.

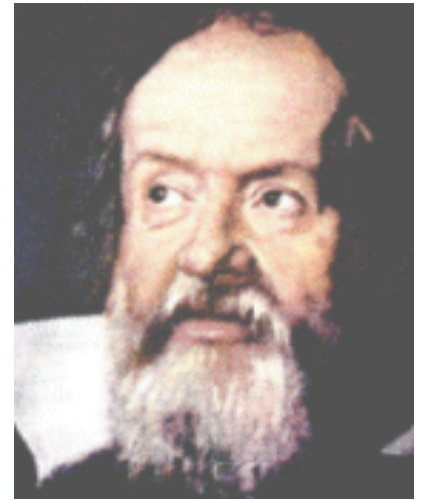


All **inertial** reference frames are **equivalent**.

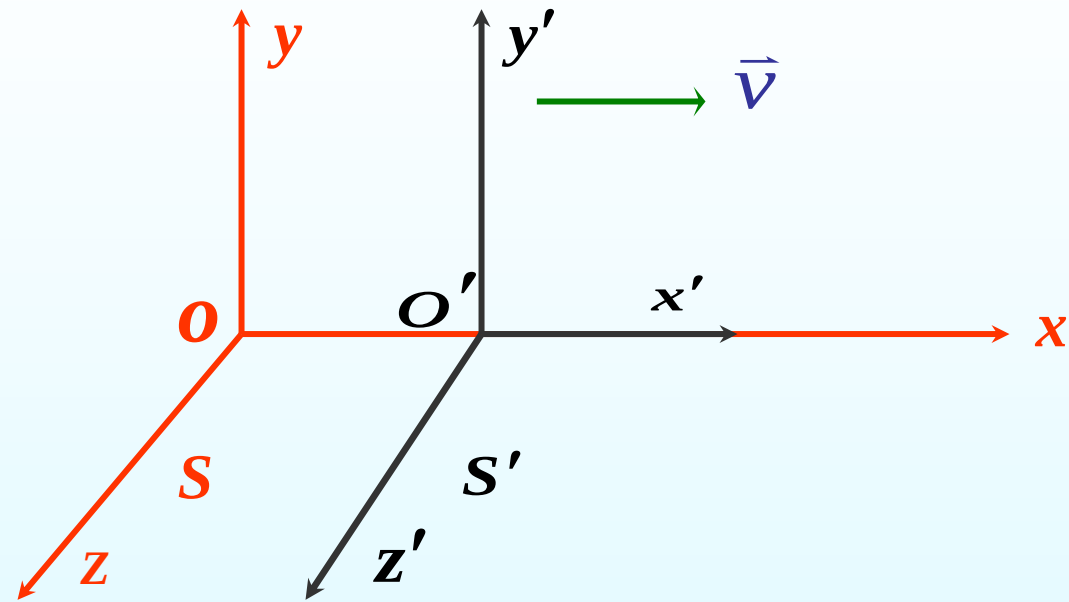
2. Galilean transformation

Think about an event in two different inertial reference frames

Reference frame S' moves to the right with respect to S at constant speed v



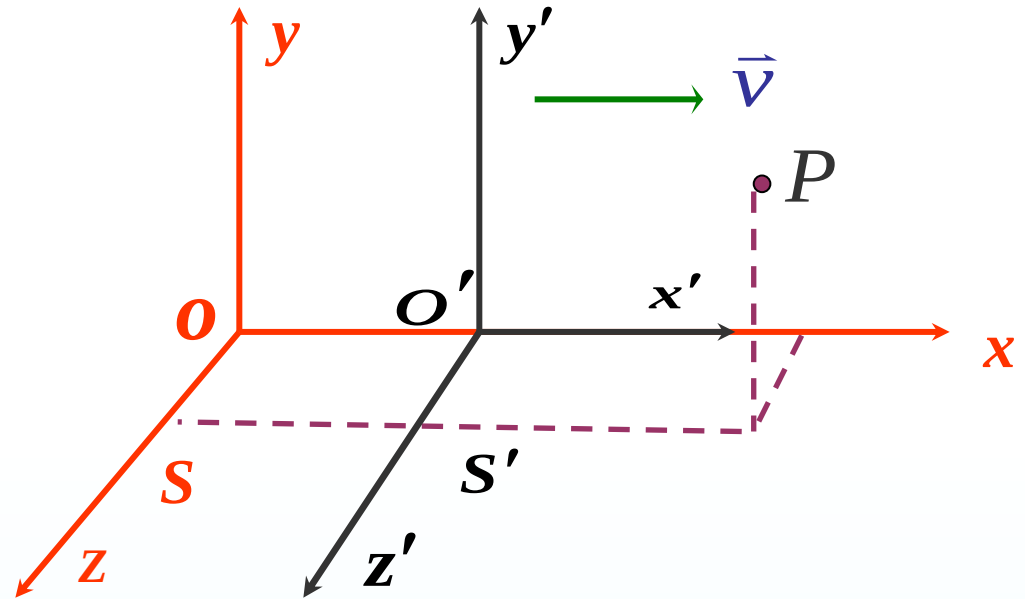
When $t = t' = 0$
the two frames are
superimposed(重合).
(标准配置)



An event occurs at some point ***P***

S frame: (x, y, z, t)

S' frame: (x', y', z', t')



•Galilean transformation equations

Transformation

$$x' = x - vt$$

$$y' = y$$

$$z' = z$$

$$t' = t$$



Inverse transformation

$$x = x' + vt'$$

$$y = y'$$

$$z = z'$$

$$t = t'$$

- Galilean velocity and acceleration transformation equations

$$\begin{aligned}x' &= x - vt \\y' &= y \\z' &= z \\t' &= t\end{aligned}$$

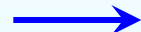


$$\begin{aligned}u'_x &= u_x - v \\u'_y &= u_y \\u'_z &= u_z\end{aligned}$$

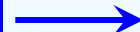


$$\begin{aligned}a'_x &= a_x \\a'_y &= a_y \\a'_z &= a_z\end{aligned}$$

$$\begin{aligned}x &= x' + vt' \\y &= y' \\z &= z' \\t &= t'\end{aligned}$$



$$\begin{aligned}u_x &= u'_x + v \\u_y &= u'_y \\u_z &= u'_z\end{aligned}$$



$$\begin{aligned}a_x &= a'_x \\a_y &= a'_y \\a_z &= a'_z\end{aligned}$$

$$\vec{a}' = \vec{a}$$

- Newton's laws satisfies the relativity principle

$$\begin{array}{ll}
 \text{Frame S:} & \vec{F} \quad m \quad \vec{a} \\
 \text{Frame S':} & \vec{F}' \quad m' \quad \vec{a}'
 \end{array}
 \left\{ \begin{array}{l} \vec{F} = m\vec{a} \\ \vec{F}' = m'\vec{a}' \end{array} \right.$$

In Newton's Mechanics

Force is **independent** of reference frame. $\vec{F} = \vec{F}'$

Mass is **independent** of motion of objects. $m = m'$

$$\because t_1 = t'_1, t_2 = t'_2, \quad \Delta t = t_2 - t_1, \Delta t' = t'_2 - t'_1$$

Time interval doesn't depend on the choice of reference frame.

$$\therefore \Delta t = \Delta t'$$

$$\Delta r = \left[(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2 \right]^{1/2}$$

$$\Delta r' = \left[(x'_2 - x'_1)^2 + (y'_2 - y'_1)^2 + (z'_2 - z'_1)^2 \right]^{1/2}$$

$$\left. \begin{aligned} x_2 - x_1 &= x'_2 - x'_1 \\ y_2 - y_1 &= y'_2 - y'_1 \\ z_2 - z_1 &= z'_2 - z'_1 \end{aligned} \right\} \rightarrow \boxed{\Delta \vec{r} = \Delta \vec{r}'}$$

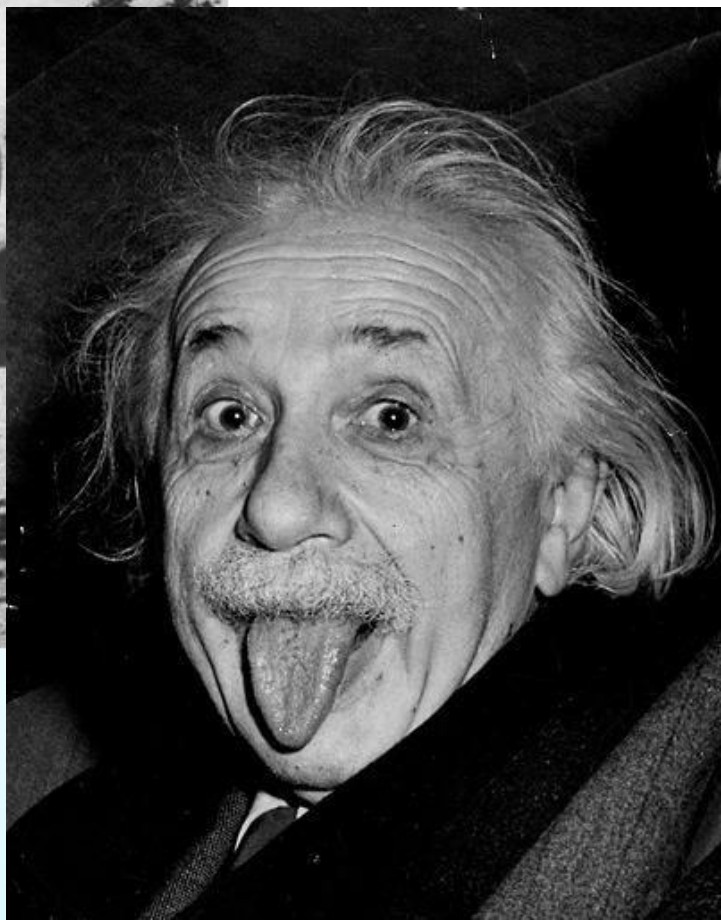
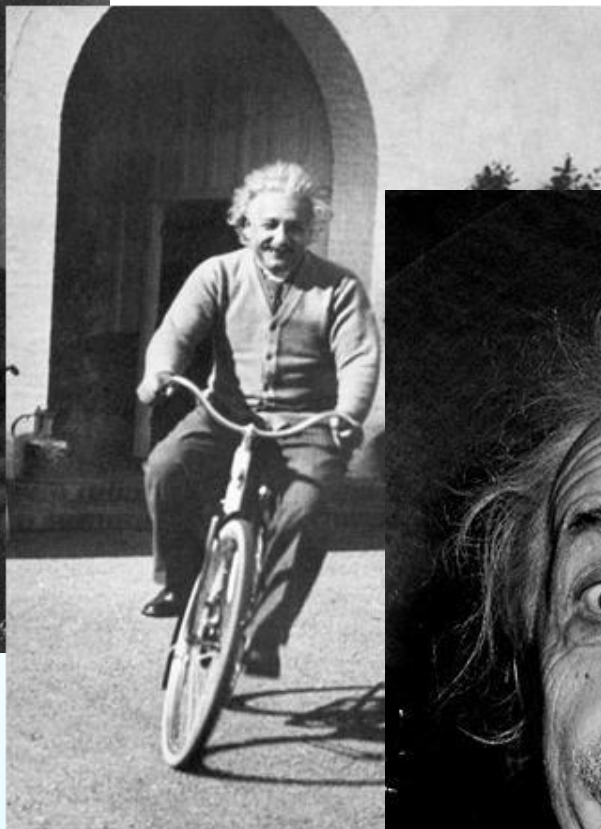
The **distance** between two points is **independent** of reference frame.

3. concepts of space and time in classical mechanics

- In classical mechanics, space and time are considered to be **absolute** and **separate**: their measurement doesn't change from one reference to another .
- The **mass** as well as all **forces** are assumed to be **unchanged** by a change in inertial reference frame.

在狭义相对论建立之前，科学家们普遍认为：**时间和空间都是绝对的**，可以脱离物质运动而存在，并且**时间与空间没有任何联系**。





21.2 Postulates of the special theory of relativity

1. 伽利略变换的困难

1) 19世纪成熟的电磁理论表明真空中光速 c 是常量。

伽利略变换：以 u 速度运动光源发出的光速不再是 c 。

2) Maxwell 方程组对伽利略变换非协变——通过电磁实验可以找到“绝对参照系”，迈克耳逊-莫雷实验的“零”结果

不同惯性系中电磁规律是否相同？

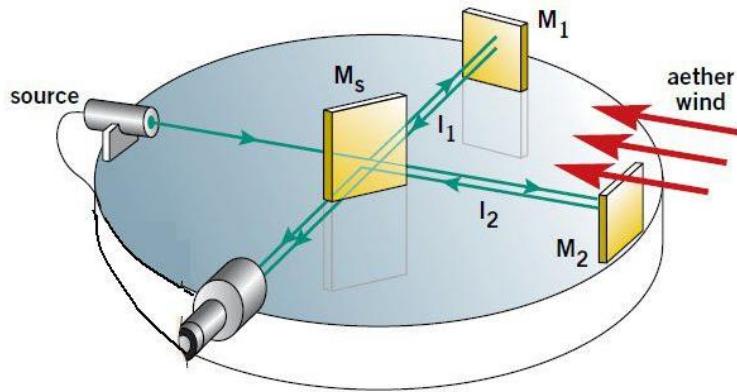
如果不同，则必然导致各惯性系不等价——应存在一个特殊的惯性系（以太），没找到！

如果相同，则坐标变换后方程组的形式应保持不变。

两种可能：{ 伽利略变换是正确的，麦克斯韦方程组必予以修正
麦克斯韦方程组是正确的，伽利略变换必予以修正

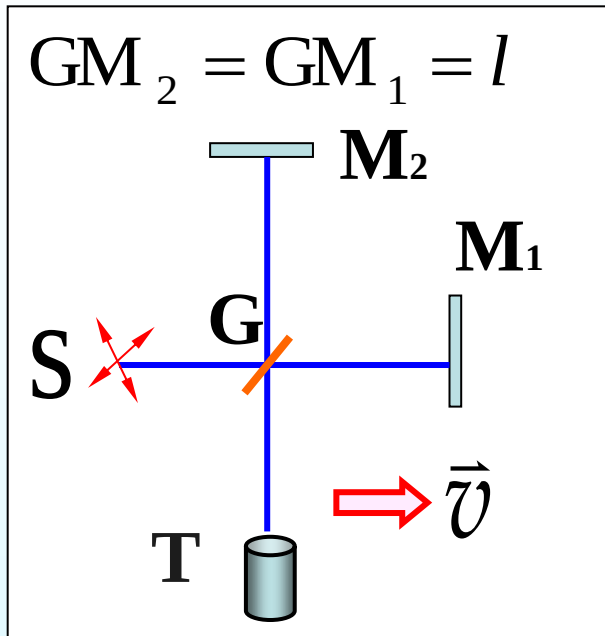
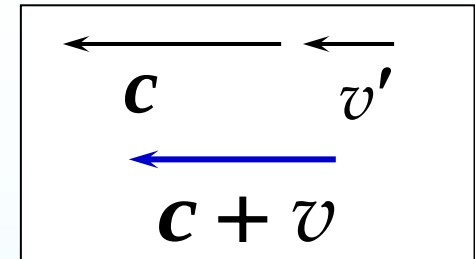
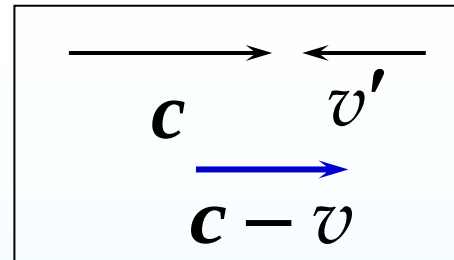


2.The Michelson-Morley Experiment



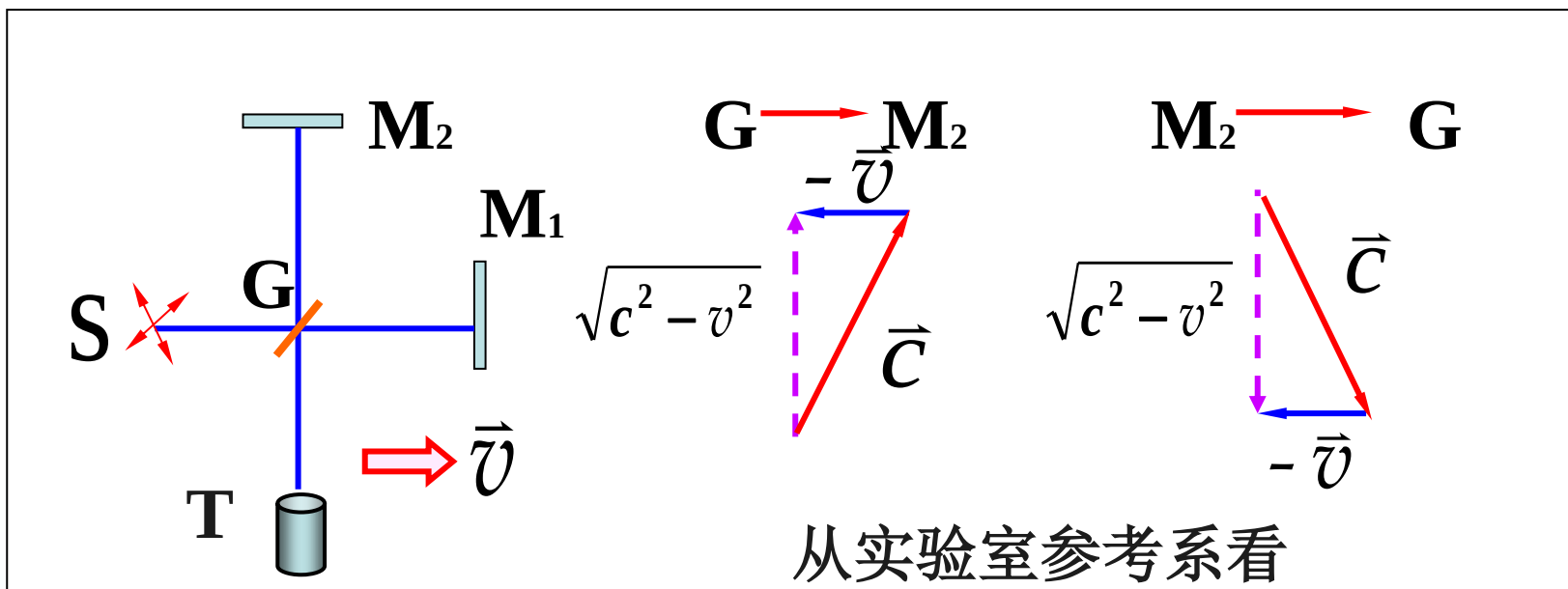
地球相对于以太速度: v
光在以太速度: c

$G \rightarrow M_1 \rightarrow G$



$$t_1 = \frac{l}{c-v} + \frac{l}{c+v} = \frac{2lc}{c^2 - v^2}$$

$$= \frac{2l}{c} \left(1 - \frac{v^2}{c^2} \right)^{-1} \approx \frac{2l}{c} \left(1 + \frac{v^2}{c^2} \right)$$



$$G \longrightarrow M_2 \longrightarrow G \quad GM_2 = GM_1 = l$$

$$t_2 = \frac{2l}{c\sqrt{1-v^2/c^2}} = \frac{2l}{c} \left(1 - \frac{v^2}{c^2}\right)^{-\frac{1}{2}} \approx \frac{2l}{c} \left(1 + \frac{1}{2} \frac{v^2}{c^2}\right)$$

$$\Delta t = t_1 - t_2 = \frac{2l}{c} \times \frac{1}{2} \frac{v^2}{c^2} = \frac{lv^2}{c^3}$$

Optical path difference: $\Delta = c(t_1 - t_2) = \frac{lv^2}{c^2}$

Rotate the instrument by 90° , the change in optical path difference

$$\Delta N = \frac{2\Delta}{\lambda} \approx 2l \frac{v^2}{\lambda c^2}$$

$$l = 10\text{m}, \lambda = 500\text{nm}, v = 3 \times 10^4 \text{m/s} \rightarrow \Delta N \approx 0.4$$

The instrument used by Michelson and Morley could detect shifts as small as 0.01 fringe .

However, it detected no shift in the fringe pattern.

Null result : $\Delta N = 0$

No significant fringes shift whatever.

Ether does not really exist.

3. Postulates of the special theory of relativity

(1) The relativity principle (相对性原理)

The laws of physics have the **same form** in all inertial reference frames.

在任何惯性系中，一切物理规律都相同，即具有相同的**数学表达式**

(2) Constancy of the speed of light(光速不变原理)

The speed of light in vacuum has the same value **c** in **all directions** and in **all inertial reference frames**.

(independent of the speed of the source or observer)

真空中的光速是常量，沿各个方向都等于 **c** ，与光源或观测者的运动状态无关。

Any inertial frame of reference can be used to describe events and do physics.

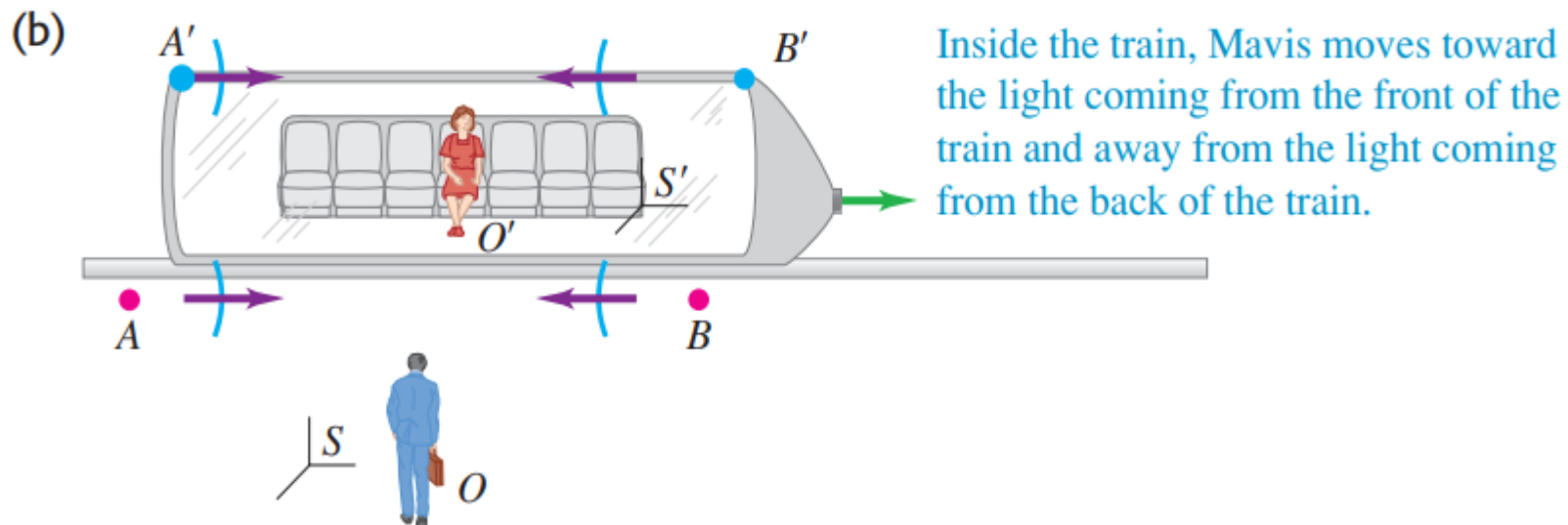
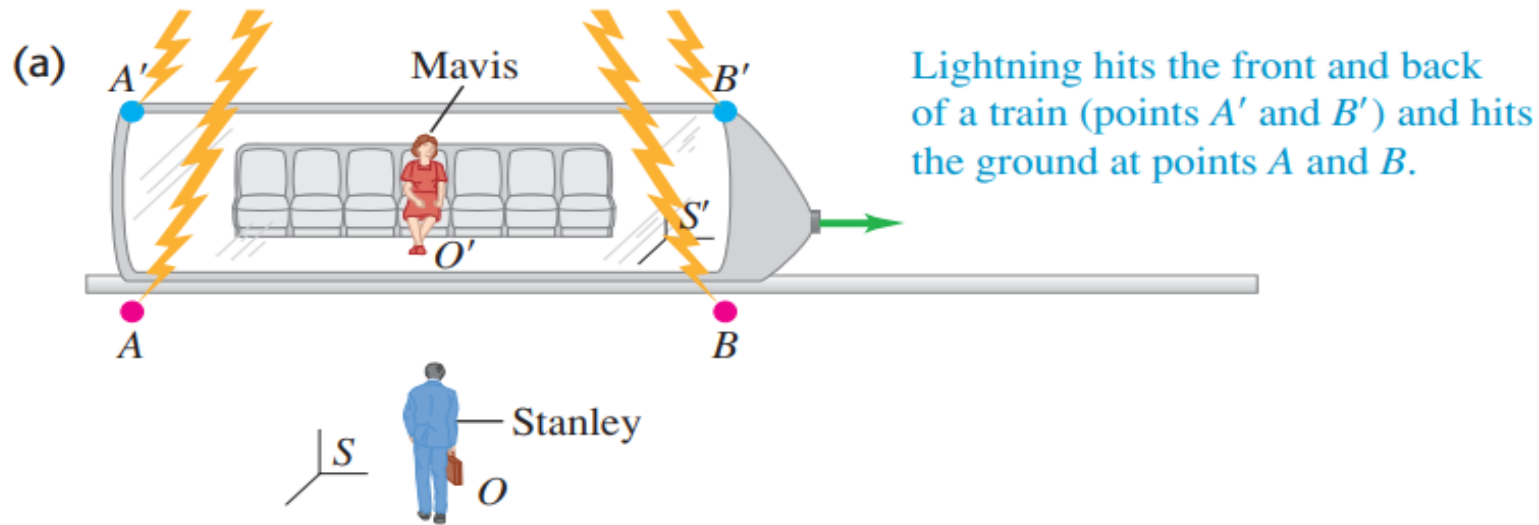
There is no preferred inertial frame of reference.

The two observers must find that light travels at the **same speed**.

Nevertheless, all observers agree on the forms of the **laws of physics** in their respective frames, because these laws must be the same for all observers in uniform motion.

21. 3 Simultaneity; Time Dilation and length Contraction

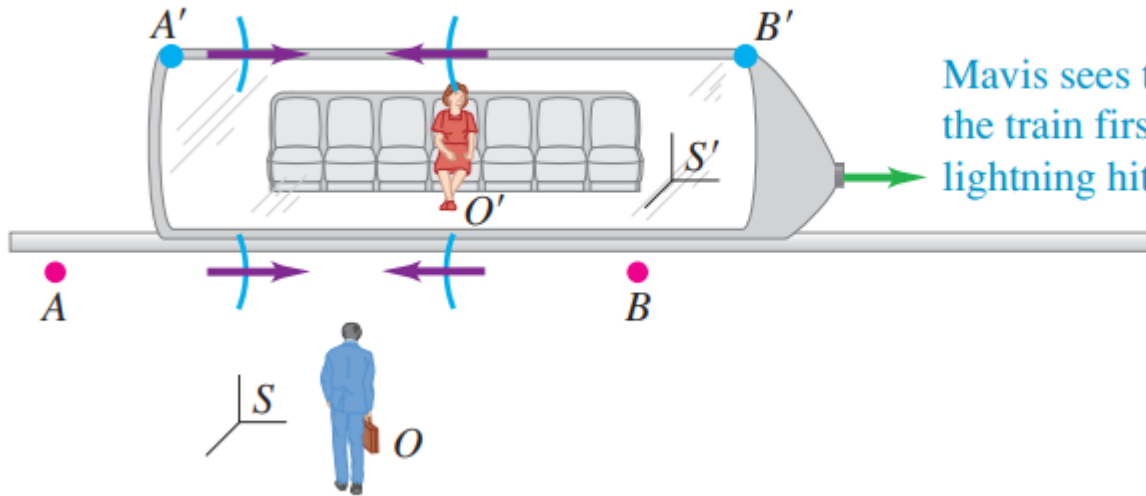
1. Simultaneity is relative(同时是相对的)



21. 3 Simultaneity; Time Dilation and length Contraction

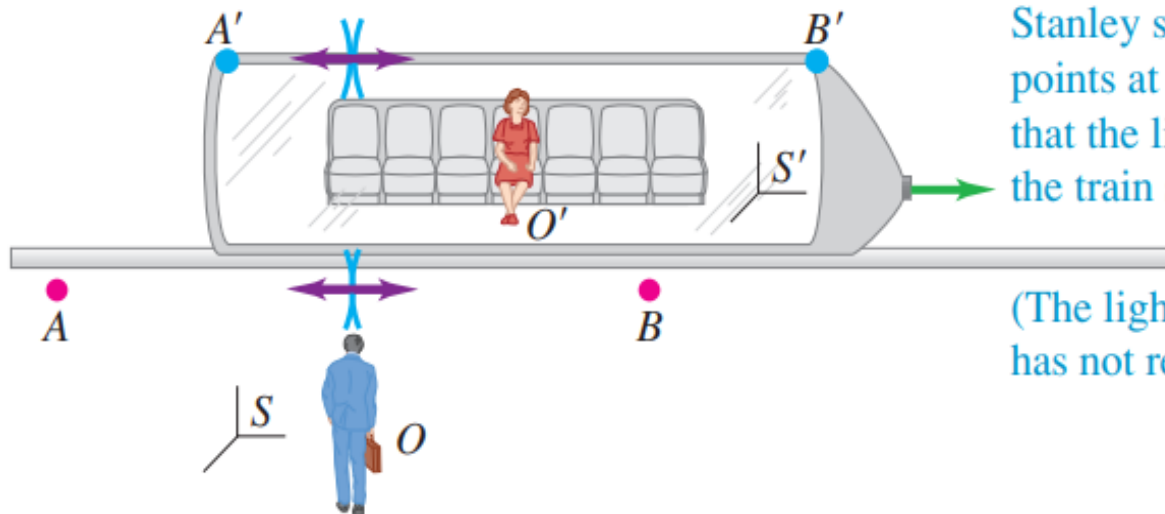
1. Simultaneity is relative

(c)



Mavis sees the light from the front of the train first, so she believes that the lightning hit the front of the train first.

(d)



Stanley sees the lightning hit the two points at the same time; so he believes that the lightning hit the two ends of the train simultaneously.

(The light from the rear of the train has not reached Mavis yet.)

Two events that are simultaneous in one reference frame are in general **not simultaneous** in a second frame moving relative to the first. That is, **simultaneity** is not an **absolute** concept but rather one that depends on the state of motion of the observer.

Measuring an event

An event is an occurrence describable by **three space coordinates** and **one time coordinate**.

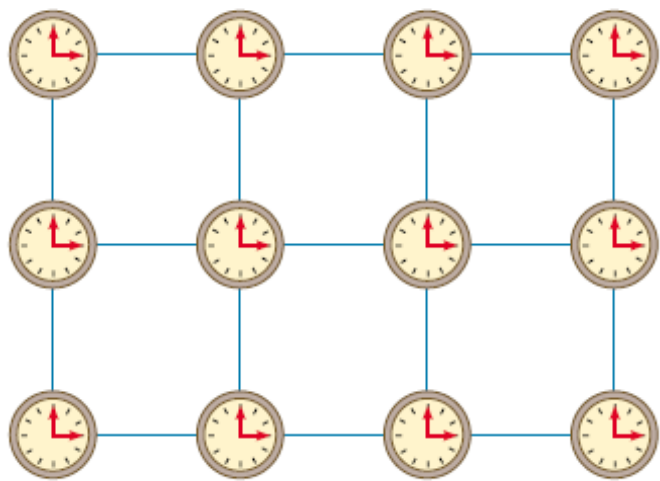
Observers in **different inertial frames** always measure different time intervals with **their clocks** and different distances with **their meter sticks**.

time interval in frame S $\Delta t = t_2 - t_1$

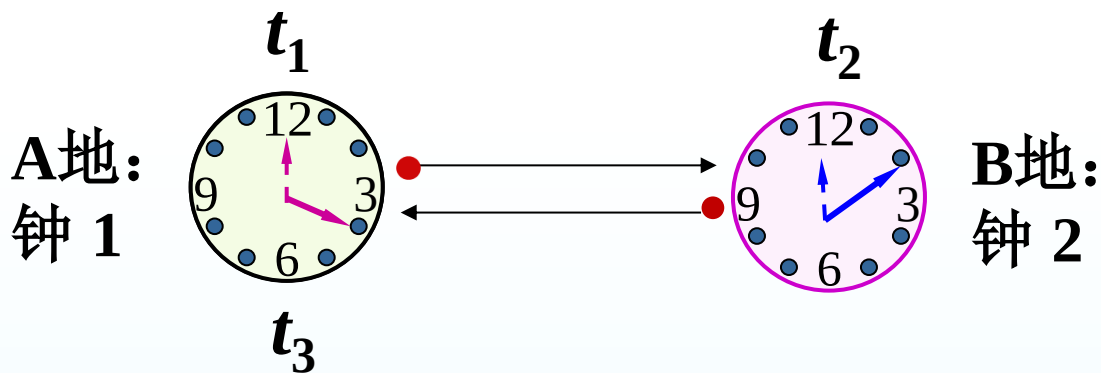
time interval in frame S' $\Delta t' = t'_2 - t'_1$

对钟 (Clock synchronization)

根据光速不变原理，用光信号把同一参考系上的不同地点所有钟校准，并使它们同步。



同一参考系不同地点钟**同步**

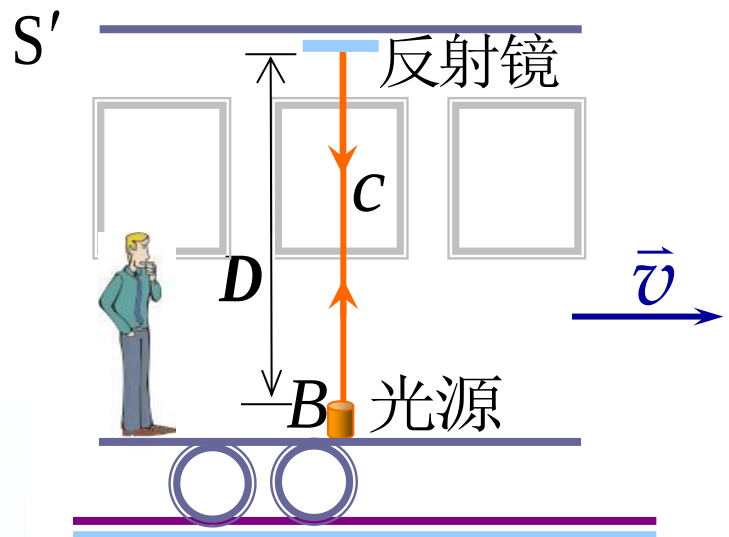


$$t_2 - t_1 = t_3 - t_2$$

两个本地钟同步

Two clock synchronization in one reference frame are **not synchronous** in any other frame moving relative to the first frame.

2. Time Dilation 动钟变慢，时间膨胀(时间延缓)

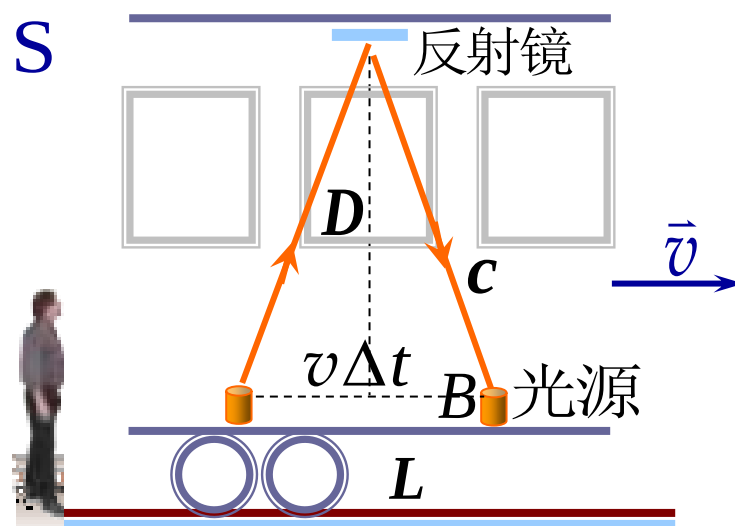


事件 1: S' 系光信号发射

S' 系 (车厢参考系)

(x'_1, y'_1, z'_1, t'_1)

(x'_2, y'_2, z'_2, t'_2)

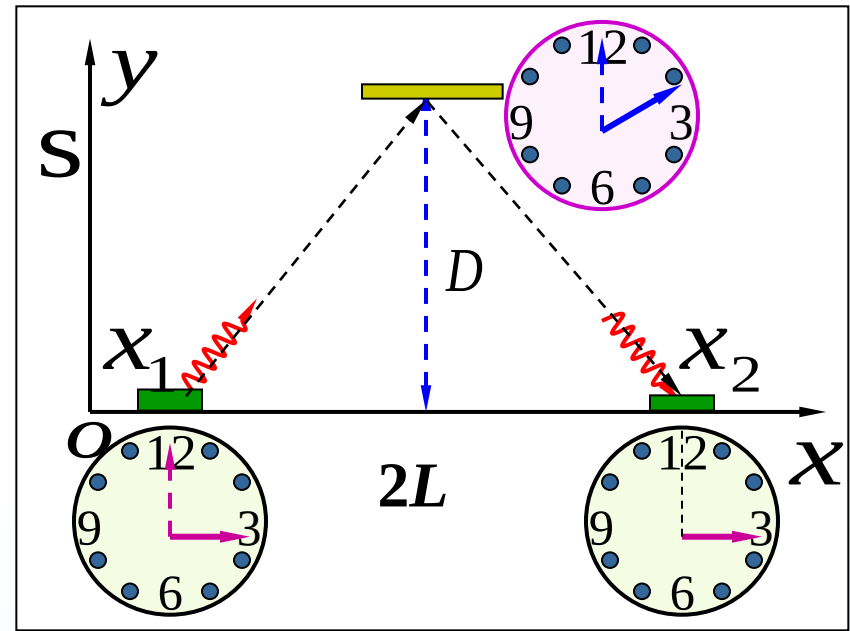
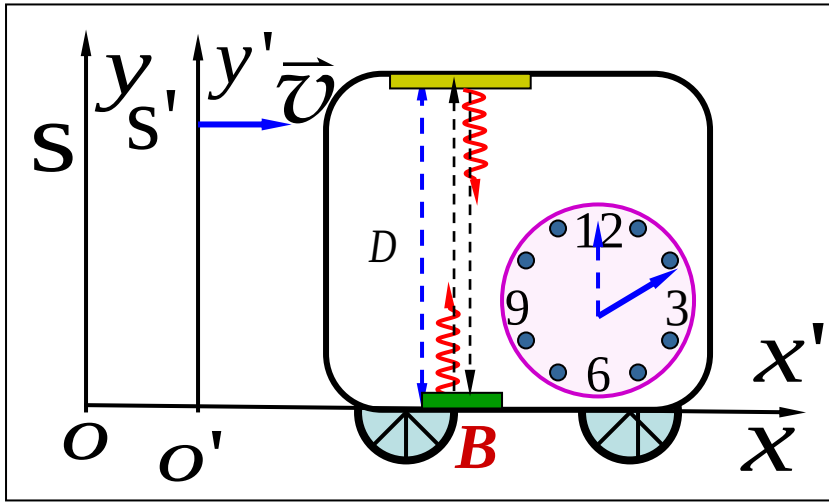


事件 2: S' 系光信号接收

S 系 (地面参考系)

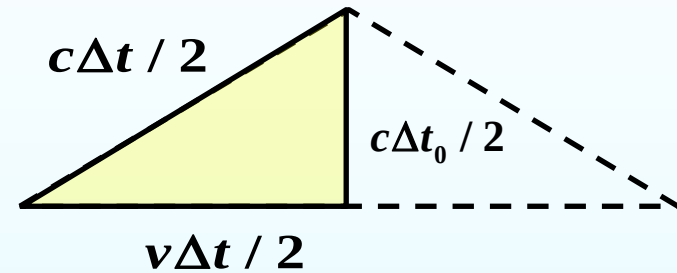
(x_1, y_1, z_1, t_1)

(x_2, y_2, z_2, t_2)



$$\Delta t_0 = \Delta t' = t'_2 - t'_1 = 2D/c$$

$$\Delta t = \frac{2\sqrt{D^2 + L^2}}{c} = \frac{2\sqrt{D^2 + \left(\frac{1}{2}v\Delta t\right)^2}}{c}$$



Eliminating D , we have

$$\Delta t = \frac{\Delta t_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$\beta = \frac{v}{c}$$

β : speed
parameter

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Time dilation factor
(Lorentz factor) 膨胀因子

$$\Delta t = \frac{\Delta t_0}{\sqrt{1 - \frac{v^2}{c^2}}} = \frac{\Delta t_0}{\sqrt{1 - \beta^2}} = \gamma \Delta t_0 > \Delta t_0 \quad \text{原时最短}$$

Proper time Δt_0 (原时、固有时间): 用一个相对事件发生地静止的钟测量的两个同地事件的时间间隔。

Δt (时间膨胀, 时间延缓): 两事件为异地事件, 需用两个钟测出其时间间隔(两地时)

Discussion

$$\Delta t = \frac{\Delta t_0}{\sqrt{1 - (v / c)^2}} > \Delta t_0$$

1. Time dilation is a relativistic effect. It means that time does not actually pass more slowly in moving reference frames, it only *seems* to pass more slowly.(时间延缓是一种相对效应)

2. **Everything** in the moving reference frame are measured to be slower.

3. If $v \ll c$, $\Delta t = \Delta t$

Example Lifetime of a moving muon.

(a) What will be the **mean lifetime** of a muon as measured in the laboratory if it is traveling at $v = 0.6c$ with respect to the lab? $\Delta t_0 = 2.2 \times 10^{-6} \text{ s}$

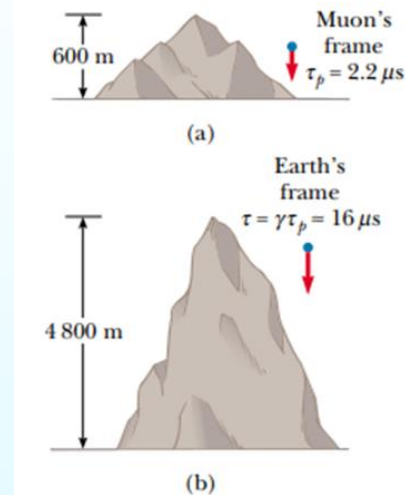
(b) How far does a muon travel in the lab?

Solution: lab : S , muon: S'

$$(a) \Delta t = \frac{\Delta t_0}{\sqrt{1 - \beta^2}} = \frac{2.2 \times 10^{-6}}{\sqrt{1 - \frac{0.6c^2}{c^2}}} = 2.8 \times 10^{-6} \text{ s}$$

$$(b) L = v\Delta t = 1.8 \times 10^8 \times 2.8 \times 10^{-6} = 500 \text{ m}$$

$$L' = v\Delta t'(\Delta t_0) = 1.8 \times 10^8 \times 2.2 \times 10^{-6} = 400 \text{ m}$$



Example Time dilation at 100km/h.

A car traveling 100km/h covers a certain distance in 10.0s according to the driver's watch. What does an observer on Earth measure for the **time**?

Solution: Earth :S , car: S' $v = 100\text{km/h} = 27.8\text{m/s}$

$$\Delta t_0 = 10.0\text{s} \quad (1 \pm x)^n \approx 1 \pm nx \quad (x \ll 1)$$

$$\Delta t = \frac{\Delta t_0}{\sqrt{1 - \beta^2}} = \frac{10.00}{\sqrt{1 - \left(\frac{27.8}{3.00 \times 10^8}\right)^2}} = \frac{10.00}{\sqrt{1 - 8.59 \times 10^{-15}}}$$

$$\Delta t = 10.00 \left[1 + \frac{1}{2} \left(\frac{27.8}{3.00 \times 10^8} \right)^2 \right] = 10.00 + 4 \times 10^{-15} \text{s}$$

半人马座 α 星(三星系统)是距离太阳系最近的恒星系统，其中比邻星(α 星C)距离地球4.24光年(ly)。设有一宇宙飞船自地球飞到比邻星，若宇宙飞船相对地球的速度为 $0.99c$ ，按地球上的时钟计算要用多少年时间？如以飞船上的时钟计算，所需时间又为多少年？

解： 地球上的时钟量测量 α 星所需时间为

$$\Delta t = \frac{L}{v} = \frac{4.24 \text{ ly}}{0.99c} = 4.28 \text{ 年}$$

若用飞船上的钟测量，飞船飞到比邻星所需时间为

$$\Delta t_0 = \Delta t \sqrt{1 - \frac{v^2}{c^2}} = \sqrt{1 - 0.99^2} \times 4.28 = 0.6 \text{ 年}$$



正是时间膨胀效应使得星际航行成为可能。

3. Length Contraction (动尺变短, 长度收缩)

The same thought can be used to illustrate that length is also a relative quantity for different frame observers.

Proper length : distance between two points are measured by **observers at rest** with respect to it .

A rod with length L_0 is still in frame S (Earth reference frame)

Earth observes

$$L_0 = v\Delta t$$



Spacecraft observes

$$L = v\Delta t_0$$

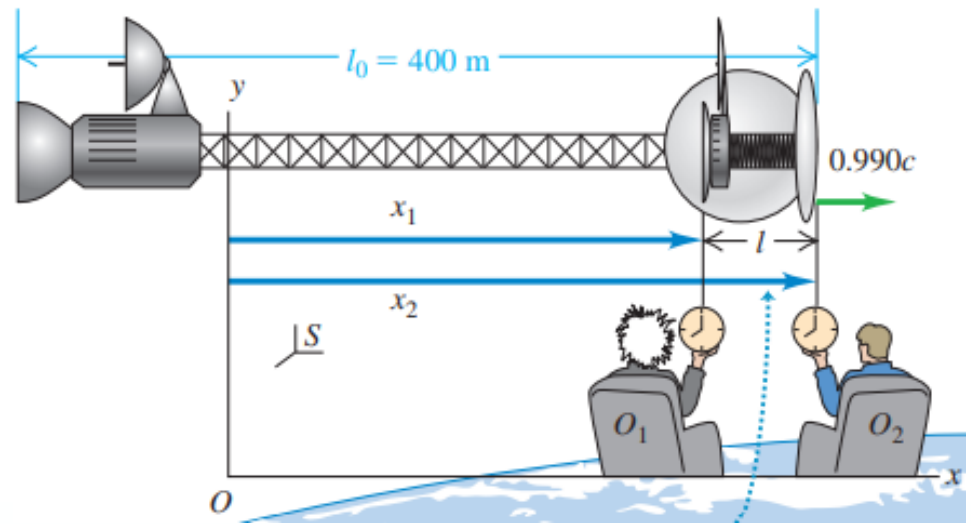
S frame

$$\Delta t = \frac{\Delta t_0}{\sqrt{1 - \beta^2}}$$

$$L = L_0 \sqrt{1 - \frac{v^2}{c^2}}$$

Discussion:

$$L = L_0 \sqrt{1 - \frac{v^2}{c^2}}$$



1. Length contraction is a **relativistic** effect.
2. Length contraction takes place only along the **direction of motion**.
3. It is important to emphasize that **proper length** and **proper time** are measured in **different reference** frames.

Example. A spaceship is traveling relative to nearby star at $0.86c$. Observers inside the ship find it takes $4 \times 10^{-8}\text{s}$ to pass a marker that is at rest relative to the star.

- (a) According to **observers in the spaceship**, what is the length of their craft?
- (b) According to **observers at rest** relative to the star, what is the length of the spaceship?

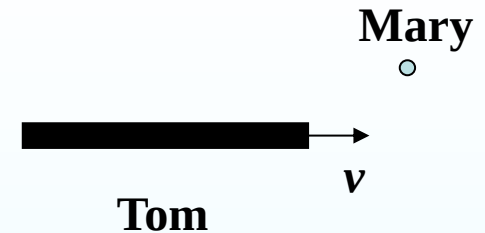
Solution:

$$(a) L_0 = v\Delta t = 0.86 \times 3 \times 10^8 \times 4 \times 10^{-8} = 10.32\text{m}$$

$$(b) L = L_0 \sqrt{1 - v^2 / c^2} = 10.32 \times \sqrt{1 - 0.86^2} = 5.27\text{m}$$

Example Mary (at point A) and **Tom's spaceship** (of proper length **$L_0=230\text{m}$**) pass each other with constant relative speed v . **Mary** measures a time interval of **$3.57\mu\text{s}$** for the ship to pass her. In terms of c , what is the **relative speed v** between Mary and the ship?

Solution: Mary: proper time
 Tom: proper length

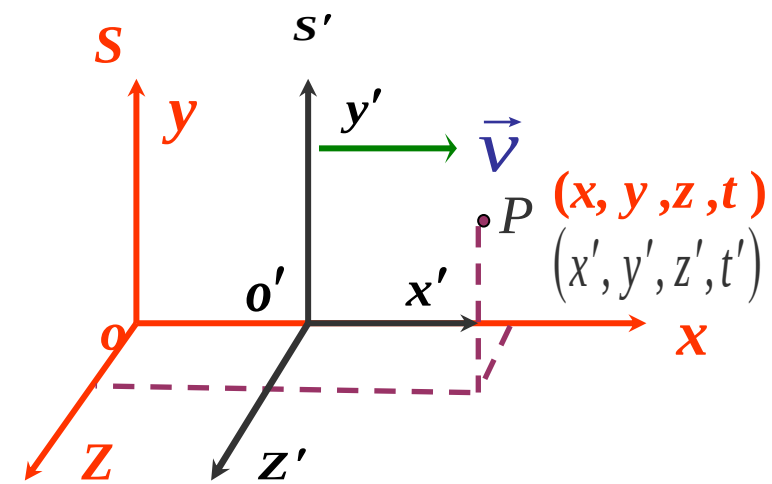


$$\Delta t_0 = \frac{L_0 \sqrt{1 - v^2 / c^2}}{v}$$

$$v = \frac{L_0 c}{\sqrt{(c \Delta t)^2 + L_0^2}} = 0.210c$$

21.4 Lorentz transformations

The same event has different space-time coordinates, $S : (x, y, z, t)$, $S' (x', y', z', t')$



Lorentz coordinate transformation

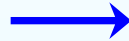
Transformation

$$x' = \frac{x - vt}{\sqrt{1 - (\frac{v}{c})^2}}$$

$$y' = y$$

$$z' = z$$

$$t' = \frac{t - \frac{vx}{c^2}}{\sqrt{1 - (\frac{v}{c})^2}}$$



Inverse Transformation

$$x = \frac{x' + vt'}{\sqrt{1 - (\frac{v}{c})^2}}$$

$$y = y'$$

$$z = z'$$

$$t = \frac{t' + \frac{vx'}{c^2}}{\sqrt{1 - (\frac{v}{c})^2}}$$

If: $\beta = \frac{v}{c}$

β : speed parameter

$$x' = \frac{x - vt}{\sqrt{1 - \beta^2}}$$

$$y' = y$$

$$z' = z$$

$$t' = \frac{t - \frac{vx}{c^2}}{\sqrt{1 - \beta^2}}$$

$$x = \frac{x' + vt'}{\sqrt{1 - \beta^2}}$$

$$y = y'$$

$$z = z'$$

$$t = \frac{t' + \frac{vx'}{c^2}}{\sqrt{1 - \beta^2}}$$

When $v \ll c$ 时, $\beta = (v/c) \ll 1$, we can get:

**Galilean
transformation**

$$x' = x - vt$$

$$y' = y$$

$$z' = z$$

$$t' = t$$

$$x = x' + vt'$$

$$y = y'$$

$$z = z'$$

$$t = t'$$

Correspondence principle (对应性原理)

• Lorentz velocity transformation

frame S $u_x = \frac{dx}{dt}, \quad u_y = \frac{dy}{dt}, \quad u_z = \frac{dz}{dt}$

frame S' $u'_x = \frac{dx'}{dt'}, \quad u'_y = \frac{dy'}{dt'}, \quad u'_z = \frac{dz'}{dt'}$

$$x' = \frac{x - vt}{\sqrt{1 - \beta^2}}$$

$$t' = \frac{t - \frac{vx}{c^2}}{\sqrt{1 - \beta^2}}$$

$$dx' = \frac{1}{\sqrt{1 - \beta^2}} (dx - v dt), \quad dt' = \frac{1}{\sqrt{1 - \beta^2}} \left(dt - \frac{v}{c^2} dx \right),$$

$$u'_x = \frac{dx'}{dt'} = \frac{dx - v dt}{dt - \frac{v}{c^2} dx} \longrightarrow$$

$$u'_x = \frac{u_x - v}{1 - \frac{v}{c^2} u_x}$$

• Lorentz velocity transformation

Transformation

$$u'_x = \frac{u_x - v}{1 - \frac{v}{c^2} u_x}$$

$$u'_y = \frac{u_y}{1 - \frac{v}{c^2} u_x} \sqrt{1 - \beta^2}$$

$$u'_z = \frac{u_z}{1 - \frac{v}{c^2} u_x} \sqrt{1 - \beta^2}$$

Inverse Transformation

$$u_x = \frac{u'_x + v}{1 + \frac{v}{c^2} u'_x}$$

$$u_y = \frac{u'_y}{1 + \frac{v}{c^2} u'_x} \sqrt{1 - \beta^2}$$

$$u_z = \frac{u'_z}{1 + \frac{v}{c^2} u'_x} \sqrt{1 - \beta^2}$$

Concepts of space and time in relativity

The relativity has changed the way we view the world. **Space and time** are **not separate** and seen to be relative. They can **intertwined** with one another and time is the fourth dimension in addition to the three dimensions of space.

Space-time coordinates: (x, y, z, t)

Space takes up three dimensions and **time is a fourth dimension**. So when we examine objects and events from different reference frames, a certain amount of space is **exchanged** for time , or vice versa.

* Lorentz transformations

The same event has different space-time coordinates, $S : (x, y, z, t)$, $S' (x', y', z')$,
 $t = t' = 0$, $x = x' = 0$.

Assuming a pair of linear relations

$$\begin{cases} x = Ax' + Bt' \\ t = Cx' + Dt' \end{cases}$$

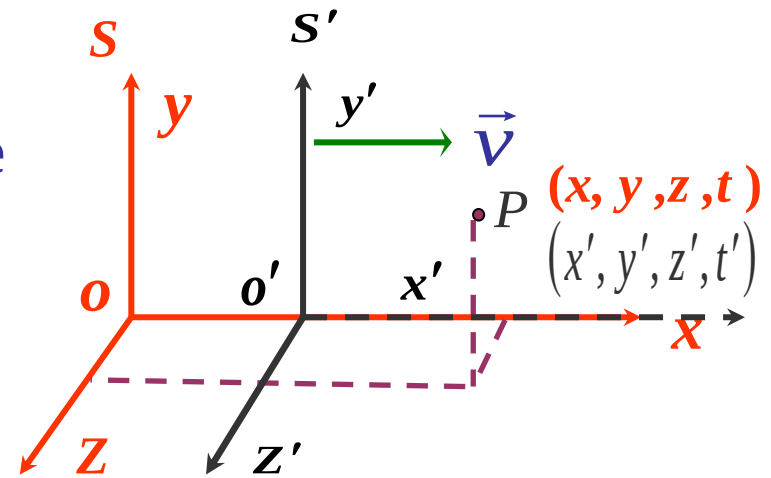
A, B, C, D can be determined in terms of v and c .

➤ The origin of S' frame is moving at constant velocity v relative to S .

$$x'=0 \rightarrow x = Bt', \quad t = Dt' \quad \frac{dx}{dt} = \frac{B}{D} = v \rightarrow B = Dv$$

➤ The origin of S frame is moving at constant velocity $-v$ relative to S' .

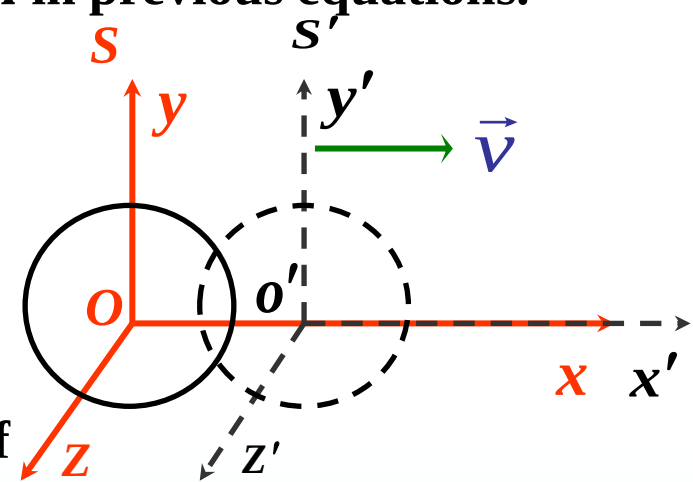
$$x=0 \rightarrow Ax' + Bt'=0 \quad \frac{dx'}{dt'} = -\frac{B}{A} = -v \rightarrow B = Av$$



We can see $D = A$, and we put $B = Av$ and $D = A$ in previous equations.

$$\begin{cases} x = A(x' + vt') \\ t = Cx' + At' \rightarrow t = A(Ex' + t') , \quad E = \frac{C}{A} \end{cases}$$

Consider a light source at rest at the origin O , and there is a stationary observer is S . At any time t after a light pulse is emitted from O , a wavefront of light spreading outward from as a sphere.



$$\begin{cases} x^2 + y^2 + z^2 = (ct)^2 \\ x'^2 + y'^2 + z'^2 = (ct')^2 \end{cases} \quad y = y', z = z' \rightarrow c^2 t^2 - x^2 = c^2 t'^2 - x'^2$$

$$c^2 A^2 (Ex' + t')^2 - A^2 (x' + vt')^2 = c^2 t'^2 - x'^2$$

$$\begin{cases} c^2 A^2 E^2 - A^2 = -1 \\ c^2 A^2 - A^2 v^2 = c^2 \rightarrow A = \frac{1}{\sqrt{1 - (\frac{v}{c})^2}} \end{cases} \quad \begin{cases} A = \frac{1}{\sqrt{1 - (\frac{v}{c})^2}} \\ B = vA \\ C = EA = (v/c^2)A \\ D = A \end{cases}$$

Discussion:

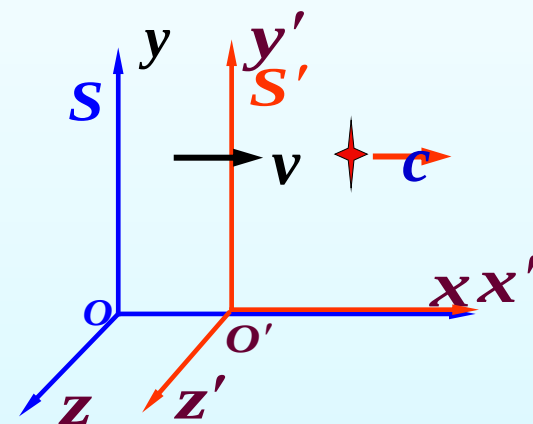
(1) 如在S系中沿x方向发射一光信号, 在S'系中观察:

$$u'_x = \frac{u_x - v}{1 - \frac{v}{c^2} u_x} = \frac{c - v}{1 - \frac{vc}{c^2}} = c$$



(2) S'系以速率c 沿x方向相对于S系运动, 如在S'系中沿x方向发射一光信号, 在S系中观察:

$$u_x = \frac{u'_x + v}{1 + \frac{v}{c^2} u'_x} = \frac{c + c}{1 + \frac{c}{c}} = c(\text{ultimate speed})$$



Time Dilation

$$t = \frac{t' + \frac{vx'}{c^2}}{\sqrt{1 - \beta^2}} = \gamma(t' + \frac{vx'}{c^2})$$

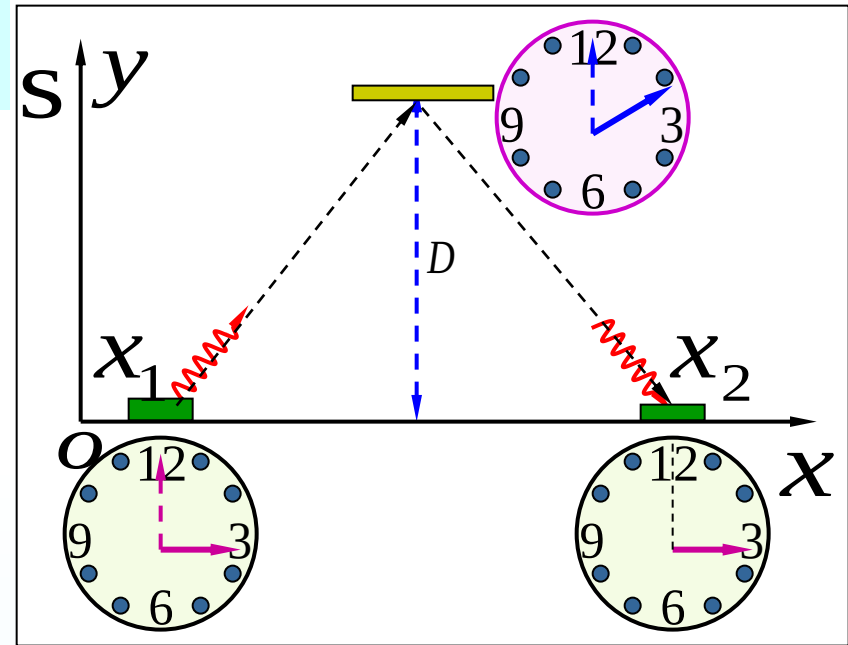
$$t_1 = \gamma(t'_1 + \frac{vx'_1}{c^2})$$

$$t_2 = \gamma(t'_2 + \frac{vx'_2}{c^2})$$

$$\Delta t = t_2 - t_1 = \gamma(\Delta t' + \frac{v\Delta x'}{c^2})$$

$$\because \Delta x' = 0 \quad \Delta t' = \Delta t_0$$

$$\Delta t = \gamma \Delta t' = \gamma \Delta t_0 = \frac{\Delta t_0}{\sqrt{1 - \beta^2}}$$



Observers in frame S

$(x_1, t_1), (x_2, t_2)$

$$\Delta t = \frac{\Delta t_0}{\sqrt{1 - \beta^2}} > \Delta t_0$$

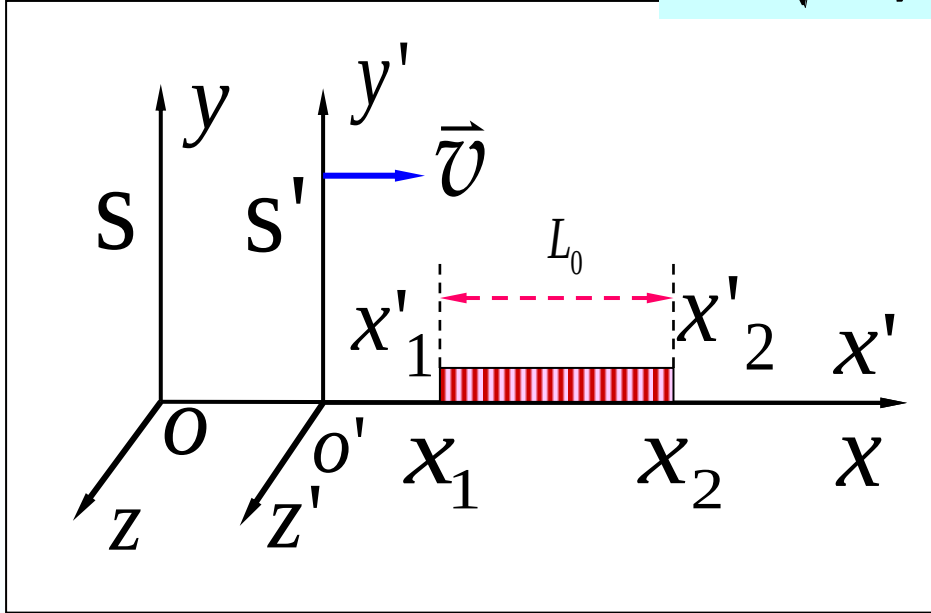
Length contraction

$$x' = \frac{x - vt}{\sqrt{1 - \beta^2}}$$

A rod with length L_0 is still in frame S'

$$L_0 = x'_2 - x'_1$$

What about the length in frame S ?



Measured the length in the **same time** $t_1 = t_2 = t$.

$$L_0 = x'_2 - x'_1 = \frac{(x_2 - vt) - (x_1 - vt)}{\sqrt{1 - \beta^2}} = \frac{x_2 - x_1}{\sqrt{1 - \beta^2}} = \frac{L}{\sqrt{1 - \beta^2}}$$

$$L = L_0 \sqrt{1 - \frac{v^2}{c^2}} < L_0$$

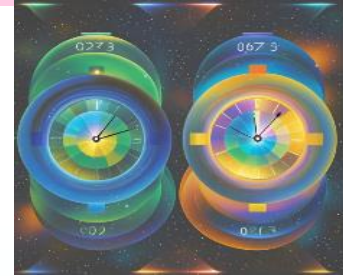
The relativity of simultaneity

$$(1) \quad \Delta x = 0 \quad \Delta t = 0 \quad \Rightarrow \Delta t' = 0$$

same place, same time ----- simultaneous

$$\Delta t' = \frac{\Delta t - \frac{v}{c^2} \Delta x}{\sqrt{1 - \beta^2}}$$

$$(2) \quad \Delta x \neq 0 \quad \Delta t = 0 \quad \Delta t' \neq 0$$



separated places, same time ----- not simultaneous

$$(3) \quad \Delta x = 0 \quad \Delta t \neq 0 \quad \Delta t' \neq 0$$

same place, not same time ----- not simultaneous

$$(4) \quad \Delta x \neq 0 \quad \Delta t \neq 0 \quad \Delta t' \neq 0$$

separated places, not same time ----- not simultaneous

$$\Delta t = \frac{v}{c^2} \Delta x \quad \text{----- simultaneous}$$

(5) If there is a causal relationship between two events, the sequence can be not reverse.(因果事件不能颠倒顺序)

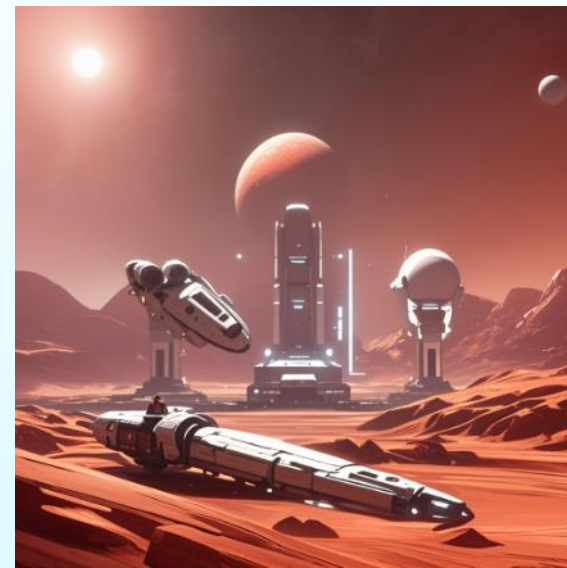
$$\Delta t' = \frac{\Delta t - \frac{v}{c^2} \Delta x}{\sqrt{1 - \beta^2}} = \frac{\Delta t}{\sqrt{1 - \beta^2}} \left(1 - \frac{v}{c^2} \frac{\Delta x}{\Delta t}\right)$$
$$= \frac{\Delta t}{\sqrt{1 - \beta^2}} \left(1 - \frac{vu}{c^2}\right)$$

$u \not\geq c, v \not\geq c;$

$\therefore \Delta t'$ must has the **same** sign with Δt .

——时序不变

*Although you cannot travel into the past,
you can travel into the future.*

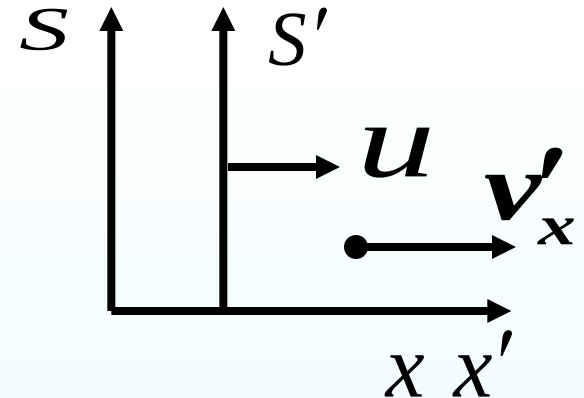


Example A particle moves along the x' axis of frame S' with speed of $0.80c$. Frame s' moves with a speed of $0.60c$ along x axis.

What is the measured **speed** of the particle in **frame S**?

Solution:

$$v = 0.60c \quad u'_x = 0.80c$$



$$u_x = \frac{u'_x + v}{1 + \frac{v}{c^2} u'_x} = \frac{0.80c + 0.60c}{1 + 0.80 \times 0.60} = 0.95c$$

Example 两个惯性系中的观测者S和S' 以0.6c的相对速度接近，在某一时刻，如果观察者S测得两者的初始距离是20m，若从此刻开始计时，试问：观察者S的测定，经过多长时间，两个观察者相遇？若按观察者S' 测定，经过多长时间，两个观察者相遇？

解 1:

$$\text{S系} \quad L_0 = 20\text{m} \quad \Delta t = \frac{L_0}{v} = \frac{20}{0.6 \times 3 \times 10^8} = 1.1 \times 10^{-7} \text{ s}$$

$$\text{S'系} \quad L = L_0 \sqrt{1 - \frac{v^2}{c^2}} = 20 \sqrt{1 - 0.6^2} = 16\text{m}$$

$$\Delta t' = \frac{L'}{v} = \frac{16}{0.6 \times 3 \times 10^8} = 8.9 \times 10^{-8} \text{ s}$$

Example 两个惯性系中的观测者S和S' 以0.6c的相对速度接近，在某一时刻，如果观察者S测得两者的初始距离是20m，若从此刻开始计时，试问：观察者S的测定，经过多长时间，两个观察者相遇？若按观察者S' 测定，经过多长时间，两个观察者相遇？

解 2:

$$\text{S系} \quad \Delta t = \frac{L_0}{v} = \frac{20}{0.6 \times 3 \times 10^8} = 1.1 \times 10^{-7} \text{ s}$$

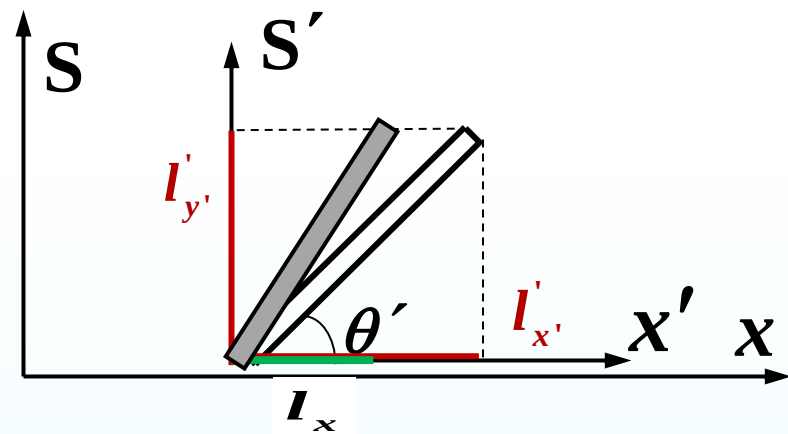
$$\text{S'系} \quad \Delta t_0 = \Delta t \sqrt{1 - \beta^2} = 8.9 \times 10^{-8} \text{ s}$$

例 一长为1 m的棒，相对于S'系静止并与 $O'x'$ 轴夹角为 θ' 。
 问：在S系的观察者来看，此棒的长度以及它与x轴的夹角为多少？（已知 $v = \frac{\sqrt{3}}{2}c$ ）

解：在 S' 系

$$\theta' = 45^\circ, \quad l' = 1 \text{ m}$$

$$l'_{x'} = l'_{y'} = \frac{\sqrt{2}}{2} \text{ m}$$



在 S 系 $l_y = l'_{y'} = \sqrt{2} / 2 m$

$$l_x = l'_{x'} \sqrt{1 - v^2 / c^2} = \sqrt{2} l' / 4$$

$$l = \sqrt{l_x^2 + l_y^2} = 0.79 \text{ m} \quad \theta = \arctan \frac{l_y}{l_x} \approx 63.43^\circ$$

例 一宇宙飞船相对地球以**0.8c** 的速度匀速飞行，一光脉冲从飞船尾传到船头，**飞船**上的观察者测得飞船长为**90m**，求：**地球**上的观察者测得**光脉冲从船尾传到船头**这两个事件的**空间间隔**是多少？

解： 飞船参考系 $\Delta x' = 90\text{m}$ $\Delta t' = \frac{\Delta x'}{c}$

地球参考系

$$\Delta x = \frac{\Delta x' + v\Delta t'}{\sqrt{1 - \frac{v^2}{c^2}}} = \frac{90 + 0.8c \frac{90}{c}}{\sqrt{1 - \left(\frac{0.8c}{c}\right)^2}} = 270\text{m}$$

$$\Delta t = \frac{\Delta x}{c} = 9 \times 10^{-7} \text{s}$$

Key words And Terms

Relativistic mass

相对论质量

Relativistic momentum

相对论动量

Rest energy

静止能量

Relativistic kinetic energy

相对论动能

Rest mass

静止质量

Mass defect

质量亏损

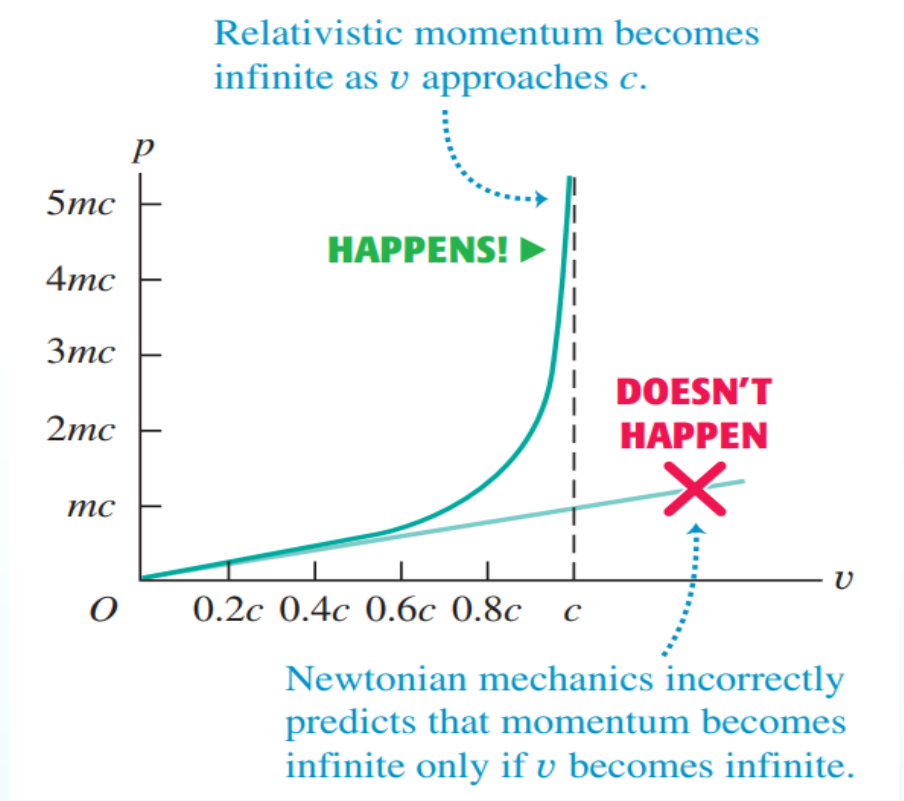
21.5 Relativistic mass, momentum and energy

1. Relativistic mass

$$m = \frac{m_0}{\sqrt{1 - \beta^2}}$$

m_0 : **rest mass** (静止质量)

The mass of an object appears to increase as its speed increases.



1901年德国物理学家考夫曼（Kaufmann）利用镭的放射性衰变中β射线的高能电子作实验，发现随速度增加，电子越来越难以加速 m 越来越大。

Discussion:

(1) The **rest** mass: Measured in a reference in which it is at rest. The rest mass of an object is an intrinsic and invariant property of the object independent of the motion of it.

(2) Mass is a **relative** quantity.

(3) If $v \ll c$, $m \rightarrow m_0$, m is a constant and will stratifies Newton's laws.

2. Relativistic momentum

$$\vec{P} = m\vec{v} = \frac{m_0 \vec{v}}{\sqrt{1 - v^2 / c^2}}$$

When $v \ll c$, $\vec{p} = m\vec{v} \rightarrow m_0 \vec{v}$

$$\vec{F} = \frac{d\vec{P}}{dt} = \frac{d(m\vec{v})}{dt} = m \frac{d\vec{v}}{dt} + \vec{v} \frac{dm}{dt} = \frac{d}{dt} \left[\frac{m_0}{\sqrt{1 - \beta^2}} \vec{v} \right]$$

The force required to change an object's momentum consists of two components, one parallel to the acceleration $d\vec{v} / dt$, due to the change in velocity and another parallel to the velocity due to change in mass.

When $v \ll c \rightarrow \vec{F} = m_0 \frac{d\vec{v}}{dt}$

3. Relativistic energy

Work-energy theorem

$$W = \int_i^f F_x dx = \int_i^f \frac{dp}{dt} dx = \int_i^f \frac{dp}{dt} v dt = \int_i^f v dp$$

$$d(pv) = p dv + v dp$$

$$W = \int_i^f d(pv) - \int_i^f p dv$$

$$\int_i^f d(pv) = pv \Big|_i^f = \frac{m_0 v^2}{\sqrt{1 - v^2 / c^2}}$$

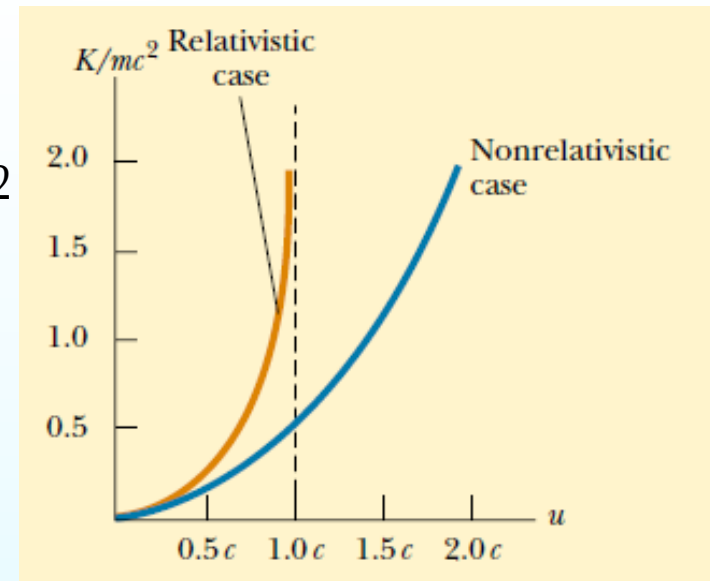
$$-\int_i^f p dv = -\int_0^v \frac{m_0 v}{\sqrt{1 - v^2 / c^2}} dv = m_0 c^2 \sqrt{1 - v^2 / c^2} \Big|_0^v = m_0 c^2 \sqrt{1 - v^2 / c^2} - m_0 c^2$$

$$W = \frac{m_0 v^2}{\sqrt{1 - v^2/c^2}} + \frac{m_0 c^2 - m_0 v^2}{\sqrt{1 - v^2/c^2}} - m_0 c^2$$

$$W = \frac{m_0 c^2}{\sqrt{1 - v^2/c^2}} - m_0 c^2 \quad W = K_f - 0$$

$$K = \frac{m_0 c^2}{\sqrt{1 - v^2/c^2}} - m_0 c^2 = \gamma m_0 c^2 - m_0 c^2$$

$$= mc^2 - m_0 c^2$$



When the mass increases from m_0 to m the quantity mc^2 changes by an amount equal to the kinetic energy required.

(1) Total energy 总能量 $E = mc^2$

(2) Rest energy 静止能量 $E = m_0c^2$

(3) Relativistic kinetic energy 相对论动能

$$K = mc^2 - m_0c^2$$

$$v \ll c, \quad K = m_0c^2 \left(1 + \frac{1}{2}v^2/c^2 - 1\right) = \frac{1}{2}m_0v^2$$

The law of **conservation of mass-energy** states that the total energy in the universe is constant. **All** forms of energy, potential, kinetic, thermal and every other kind of energy contribute to the relativistic mass of a body.

Discussion

$$E = mc^2$$

Relation between mass and energy

$$\Delta E = (\Delta m)c^2$$

Mass defect

Mass and energy are **equivalent** quantities, related by a constant of conversion. The equation is one of the most important discoveries in twentieth century physics.

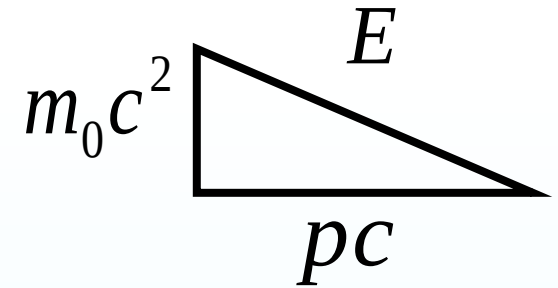
相对论的质能关系为开创原子能时代提供了理论基础，这是一个具有划时代意义的理论公式。



4. Relation between the total energy E and momentum

$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}} \rightarrow m \sqrt{1 - \frac{v^2}{c^2}} = m_0 \quad \text{Square the equation and times } c^4$$

$$m^2 c^4 = m^2 v^2 c^2 + m_0^2 c^4 \quad \left\{ \begin{array}{l} p = mv \\ E = m c^2 \\ E_0 = m_0 c^2 \end{array} \right.$$



$$E^2 = p^2 c^2 + E_0^2$$

momentum $p = \frac{1}{c} \sqrt{E^2 - E_0^2} = \frac{1}{c} \sqrt{K(K + 2E_0)}$

$$= c \sqrt{m^2 - m_0^2} = m_0 c \sqrt{\gamma^2 - 1}$$

Question: A particle with rest mass m_0 moves with speed $0.6c$. Its kinetic energy is:

A. $0.18m_0c^2$

B. $0.22m_0c^2$

C. $0.25m_0c^2$

D. m_0c^2

E. $1.25m_0c^2$

Example An electron(电子) has a **total energy** that is **five times** its **rest energy**. Determine (a) its speed and (b) its momentum.

Solution: (a) $E = mc^2 = \frac{m_0 c^2}{\sqrt{1 - v^2/c^2}} = 5m_0 c^2$

$$1 - v^2/c^2 = \frac{1}{25}, \quad v = \frac{\sqrt{24}}{5}c = 2.94 \times 10^8 \text{ m/s}$$

(b) $p = mv = \frac{m_0 v}{\sqrt{1 - v^2/c^2}} = 5m_0 v$

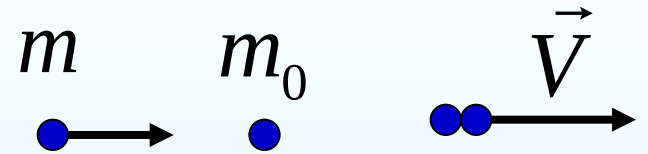
$$= 5 \times 9.1 \times 10^{-31} \times 2.94 \times 10^8 = 1.34 \times 10^{-21} (\text{kg} \cdot \text{m/s})$$

$$E^2 = p^2 c^2 + E_0^2$$

$$p = \frac{\sqrt{E^2 - E_0^2}}{c} = \sqrt{24} m_0 c = 1.34 \times 10^{-21} (\text{kg} \cdot \text{m/s})$$

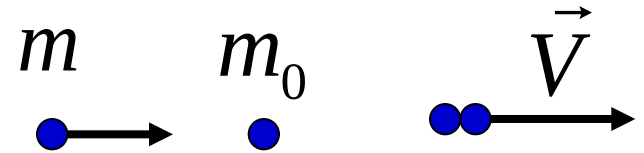
Example A particle of **rest mass** m_0 and **kinetic energy** $5m_0c^2$ strikes a **stationary** particle of **rest mass** m_0 . The resulting composite particle has **rest mass** M_0 and moves with a **speed** V , as shown. Determine: (a) What is the **total momentum** of the system before the collision? (b) What is the **total energy** of the system before the collision? (c) What is the **speed** V of the composite particle? (d) What is the **rest mass** M_0 in terms of m_0 ?

Solution:



$$(a) P = P_A = \frac{1}{c} \sqrt{E_A^2 - E_{A0}^2} = \frac{1}{c} \sqrt{36m_0^2c^4 - m_0^2c^4} = \sqrt{35}m_0c$$

$$(b) E = E_A + E_B = m_0c^2 + 5m_0c^2 + m_0c^2 = 7m_0c^2$$



Conservation of energy and momentum

$$(c) p' = p = MV = \sqrt{35}m_0c \quad E' = E = Mc^2 = 7m_0c^2$$

$$V = \frac{p'c^2}{E'} = \frac{\sqrt{35}m_0c}{7m_0c^2} c^2 = 0.85c$$

$$(d) E_0 = M_0c^2 = \sqrt{E'^2 - (p'c^2)^2}$$

$$= \sqrt{49m_0^2c^4 - 35m_0^2c^4} = \sqrt{14}m_0c^2$$

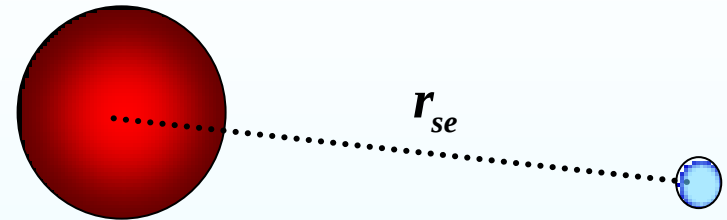
$$M_0 = \sqrt{14}m_0$$

Example Mass defect

Radiation from the sun reaches the earth at a rate of **1400 W/m^2** at the equator at noon on the first day of spring.

- (a) What is the **loss in mass** of the sun each second?
- (b) What is the **percentage decrease** in the mass of the sun over 10 billion year? (The distance from the earth to the sun is $1.5 \times 10^{11} \text{ m}$ and the present mass of the sun is $2.0 \times 10^{30} \text{ kg}$.)

Solution :

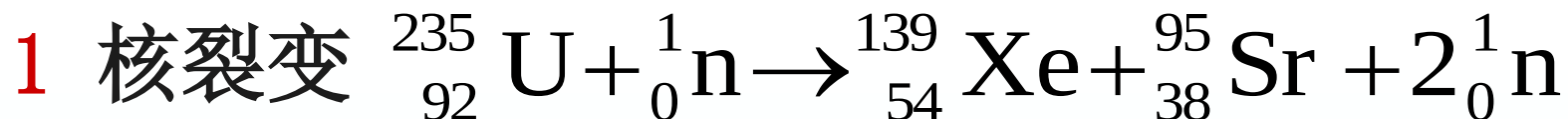


$$(a) P = 4\pi r_{se}^2 I = 4 \times 3.14 \times (1.5 \times 10^{11})^2 \times 1400 = 3.96 \times 10^{26} \text{ W}$$

$$(b) \frac{\Delta m}{m} = \frac{\Delta E}{E} = \frac{\Delta E}{mc^2} = \frac{3.96 \times 10^{26} \times 3.154 \times 10^7 \times 10^{10}}{2.0 \times 10^{30} \times (3 \times 10^8)^2} = 0.069\%$$

质能公式在原子核裂变和聚变中的应用

重核裂变、轻核聚变所释放的能量是以质量减少为代价而获得的能量，这一现象称为“质量亏损”



质量亏损 $\Delta m = 0.22\text{u}$

原子质量单位 $1\text{u} = 1.66 \times 10^{-27} \text{ kg}$

放出的能量 $Q = \Delta E = \Delta m \cdot c^2 \approx 200 \text{ MeV}$

1g 铀—235 的原子裂变释放的能量

$$Q = 8.5 \times 10^{10} \text{ J}$$

fission



原子弹爆炸（核裂变）

2 轻核聚变 ${}^2_1\text{H} + {}^2_1\text{H} \rightarrow {}^4_2\text{He}$

氘核 $m_0({}^2_1\text{H}) = 3.3437 \times 10^{-27} \text{ kg}$

氦核 $m_0({}^4_2\text{He}) = 6.6425 \times 10^{-27} \text{ kg}$

质量亏损 $\Delta m = 0.026 \text{ u} = 4.3 \times 10^{-29} \text{ kg}$

释放能量 $Q = \Delta E = (\Delta m)c^2 = 3.87 \times 10^{-12} \text{ J}$

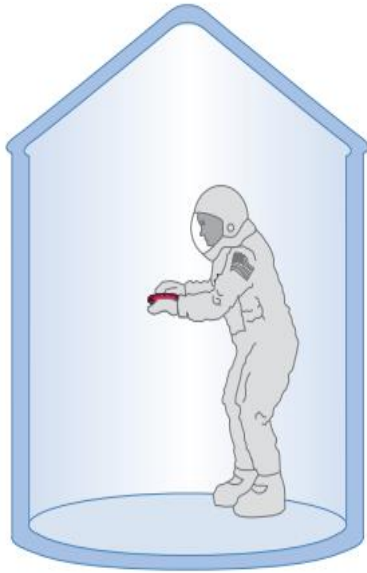
轻核聚变**条件** 温度达到 10^8 K 时，
使 ${}^2_1\text{H}$ 具有 10 keV 的动能，足以克服
两 ${}^2_1\text{H}$ 之间的库仑排斥力。

fusion

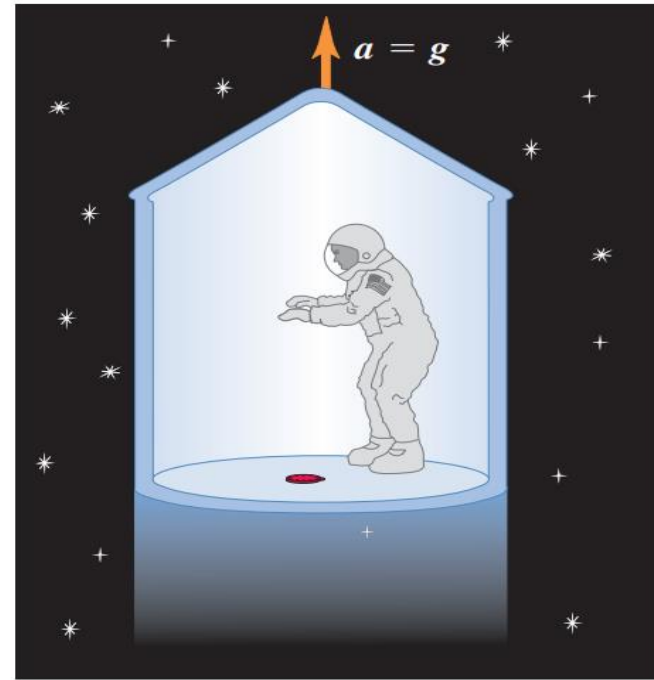


氢弹爆炸（核聚变）

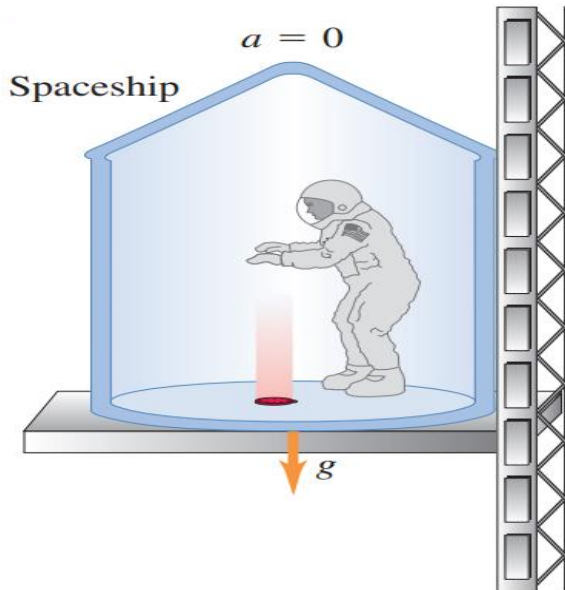
(a) An astronaut is about to drop her watch in a spaceship.



(b) In gravity-free space, the floor accelerates upward at $a = g$ and hits the watch.

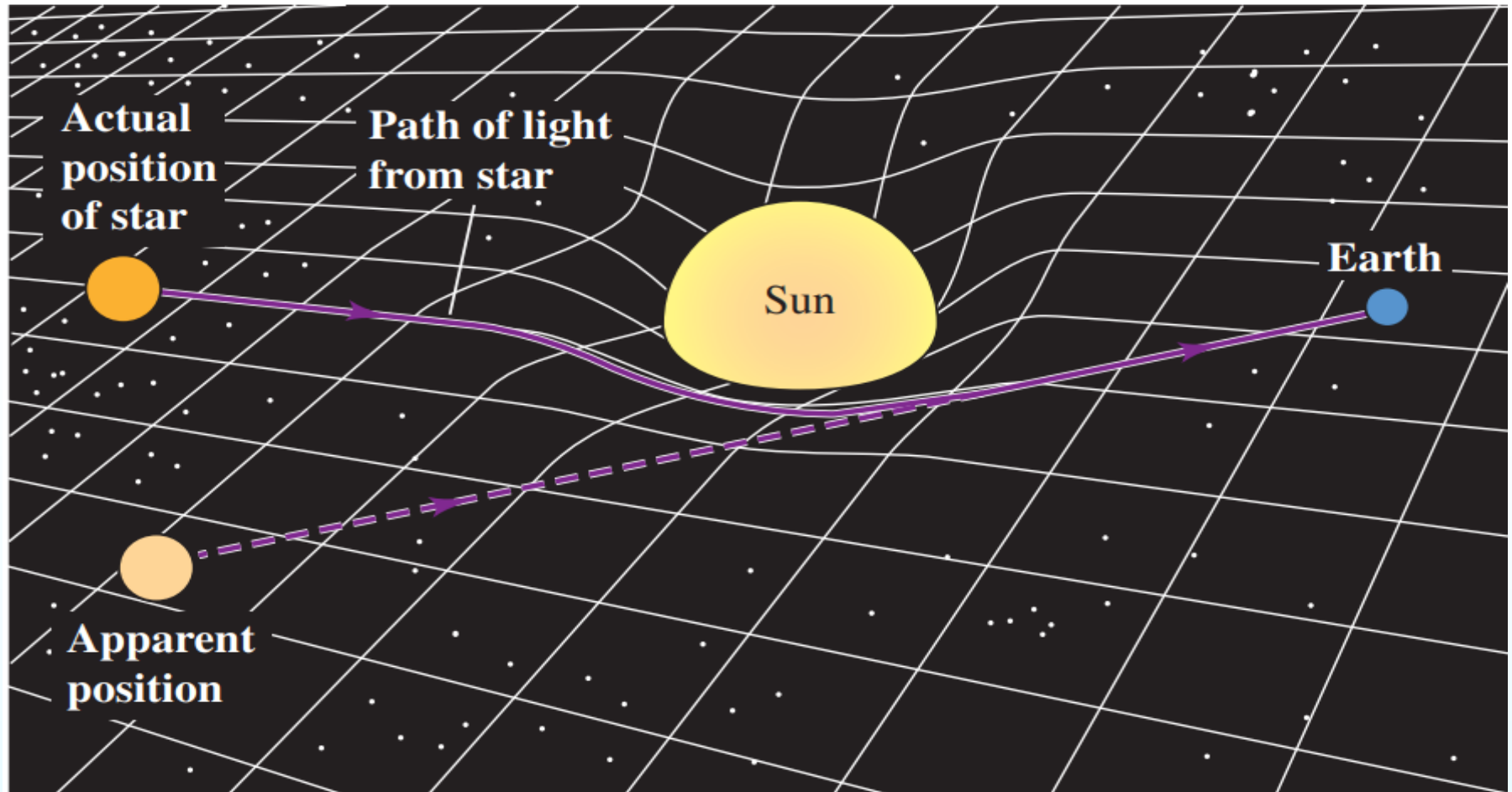


(c) On the earth's surface, the watch accelerates downward at $a = g$ and hits the floor.



If we **cannot** distinguish experimentally between a **uniform gravitational field** at a particular location and a **uniformly accelerated reference frame**, then there cannot be any real distinction between the two.

Mass tells **space** how curve; **space** tells mass how **move**.
----- J.A.Wheeler

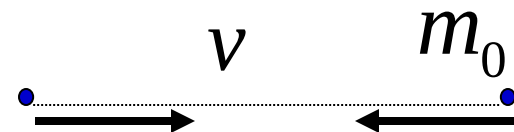


We may try to represent any gravitational field in terms of special characteristics of the coordinate system. In the general theory of relativity the geometric properties of space are affected by the presence of matter.

两全同粒子（静止质量 m_0 ）以相同的速率 v 相向运动，碰后合在一起生成一个新粒子。

求：复合粒子的速度和质量

解：设复合粒子质量为 M 速度为 $\vec{V}$



动量守恒 $m v + (-mv) = MV \Rightarrow V = 0$

由能量守恒 $2mc^2 = M_0c^2$

$$M_0 = 2m = \frac{2m_0}{\sqrt{1 - \frac{v^2}{c^2}}} > 2m_0$$

损失的能量转换成静止能量

一原子核相对于实验室以 $0.6c$ 运动，在运动方向上向前发射一电子，电子相对于核的速度为 $0.80c$ ，当实验室中测量时，求

(1) 电子的速率，(2) 电子质量 (3) 电子动能 (4) 电子动量的大小

解：S 系：实验室，S' 系：原子核

$$(1) v = 0.6c, u'_x = 0.8c$$

$$u_x = \frac{u'_x + v}{1 + \frac{v}{c^2} u'_x} = \frac{0.6c + 0.8c}{1 + \frac{0.6c \times 0.8c}{c^2}} = \frac{35}{37} c = 0.946c$$

$$(2) m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}} = \frac{m_0}{\sqrt{1 - \frac{35^2}{37^2}}} = \frac{37}{12} m_0$$

$$(3) K = E - E_0 = mc^2 - m_0c^2 = \frac{37}{12}m_0c^2 - m_0c^2 = \frac{25}{12}m_0c^2$$

$$(4) P = mv = \frac{37}{12}m_0 \times \frac{35}{37}c = \frac{35}{12}m_0c$$

$$\text{or } P = \frac{1}{c} \sqrt{E^2 - E_0^2} = \frac{1}{c} \sqrt{(E - E_0)(E + E_0)} = \frac{35}{12}m_0c$$

例：一粒子的静止质量为 m_0 ，当其动能等于其静止能量时，求其质量、速率、和动量。

$$K = E - E_0 = mc^2 - m_0c^2 = m_0c^2$$

$$mc^2 = 2m_0c^2 \rightarrow m = 2m_0$$

$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}} \quad 2m_0 = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}} \rightarrow v = \frac{\sqrt{3}}{2}c$$

$$p = mv = 2m_0 \times \frac{\sqrt{3}}{2}c = \sqrt{3}m_0c$$

$$p = \frac{1}{c} \sqrt{E^2 - E_0^2} = \frac{1}{c} \sqrt{(E - E_0)(E + E_0)} = \frac{1}{c} \sqrt{K(K + 2E_0)} = \sqrt{3}m_0c$$